

DFS – Depth First Search:

Definition:

DFS is a graph/tree traversal algorithm that explores as deep as possible before backtracking. It uses a Stack or Recursion.

Algorithm:

1. Start from the given node.
2. Mark the node as visited.
3. Visit an unvisited adjacent node.
4. Continue going deeper.
5. If no unvisited node is available, backtrack.
6. Repeat until all reachable nodes are visited.

Out put:

```
Enter number of vertices: 5
Enter number of edges: 5
Enter edges (u v):
0 1
0 2
1 3
1 4
2 4
Enter starting vertex: 0
DFS Traversal: [0, 1, 3, 4, 2]
Execution Time: 2.939300611615181e-05 seconds
Time Complexity: O(V + E)

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity: $O(V + E)$

Space Complexity: $O(V)$